

Going European from American: Does Style Matter?

Rajesh Pathak, Satish Kumar¹ and Ranajee

*Department of Finance, IBS Hyderabad, ICFAI Foundation for Higher Education,
Survey Number 156/157, Dontanpally Village, Shankarpally Mandal, R.R. District,
Hyderabad, Telangana, India (501203)*

Abstract

We examine the cross sectional predictive power and the information content of volatility smirks for future stock returns using single stock options in a setup when options settlement style is changed from American to European. The volatility smirk of European style options, unlike American style options, predict the underlying cross-sectional equity returns. Firms with steepest volatility smirk underperform firms with flatter volatility smirks, by an average of 3.28% and 4.01% annually for American and European options respectively. The results confirm that a trader with negative information prefers to trade out-of-the-money put options. The more pronounced results of European options designate the trader's preference to less risky European style stock options. Results are robust and signify the delay of equity market in incorporating information impounded in volatility smirk.

Keywords: European; American; Stock Options; Volatility Smirk; Price Discovery

JEL Classification: G11, G12, G14

¹ Corresponding author. Email ID: satishwar@ibsindia.org
Tel.: +91-9951707097

1. INTRODUCTION

Information diffusion in asset prices is one of the most important concerns of market participants when homogeneous assets trade on more than one market. Due to the distinct microstructure of markets, an informed trader prefers to trade in markets that magnify his trading incentives and makes these markets leader in price discovery (Xing, Zhang and Zhao, 2010). Derivatives markets are found to be the venue of informed trading in numerous studies² examining the direction of information flow between spot and derivatives markets across time and space. One of the major anomalies reported in the literature of derivatives is the presence of prominent volatility smirk implicit in the prices of options (e.g., see, Bates, 2003 and Xing et al., 2010). Volatility smirk refers to the difference of the implied volatilities of out-of-the-money (OTM) put options and at-the-money (ATM) call options for the same underlying. The shape and patterns of volatility smirk and reasons thereof have been analysed well for both index options (Bates, 1991; Pan, 2002 and Chen and Xu, 2014) and stock options (Bollen and Whaley, 2004 and Xing et al., 2010). However, the utility of volatility smirk in enhancing the incentives of traders remains to be examined in detail.

The presence of volatility smirk in the prices of index and stock options violates the Black–Scholes (1973) options pricing theory. The theory suggests that every option should imply the same volatility for a given underlying. However, the presence of volatility smirk specifies that implied volatility as a function of exercise/moneyness for a certain maturity is a negatively skewed curve for most of the options (Foresi and Wu, 2005). The recent options pricing theories (e.g., see Bates, 1991; Pan, 2002; and Kim and Zhang, 2013) suggest that the presence of volatility smirk could be attributed to the aversion of investors to future negative jumps in the prices of the underlyings.

In addition, series of options prices provide insights into the climate of expectations of various market participants. For instance, a call option pays off only when it finishes in-the-money (ITM) implying that the underlying asset price is in excess of the exercise price. On the other hand, a put option pays off only when the reverse is true. Thus, the set of call and put option prices across all exercise prices provides a very direct indication of market participants' aggregate subjective distributions (Bates, 1991). Therefore, an assessed downside risk in the market will lead 'put options with exercise prices well below the current spot price' (OTM puts), being priced higher than 'calls with exercise prices close to the spot

² For instance, see, Jong and Donders, (1998); Chen et. al. (2005); Ni et. al. (2008), Kumar and Tse (2009), Wang (2011), Theissen (2011), Cummings and Frino (2011); to mention a few.

price' (ATM calls). Large downward movements in the market make the 'puts' more likely to finish ITM than the 'calls' and results in steeper volatility smirk. Pan (2002) reports that investors tend to choose OTM puts to express their concerns about possible future negative jumps making OTM puts more expensive. Xing et al. (2010) report a significant predictive power of volatility smirk for the future stock returns.

Given the prevalence of volatility smirk, the primary objective of our paper is to explore whether this anomaly predicts future returns and results in profitable opportunities. Furthermore, we intend to capture an important phenomenon on National Stock Exchange of India (NSE) when the American trading style on stock options is changed to the European trading style. Such a change in options trading style should affect the informativeness of options market due to inherent risk characteristics of European options. The next important objective of our paper is to examine the information content of volatility smirks in predicting future stock returns using the portfolio approach.

We contribute to the literature by examining the inter-linkage of spot and options market at firm level in a unique market setup. An impressive range of researchers³ examine such inter-linkages across markets at firm level and provide mixed evidence of information content in prices and trading activity of the options market. However, our paper differs from existing studies on several grounds. *One*, most of the studies examine information content of options prices or volume (Chan et al., 2002; Pan and Poteshman, 2006) about future stock returns; not many studies examine the information content of volatility smirk using single stock options. More particularly, no such study exists for the Indian market. *Two*, we examine the information content of volatility smirk in a unique market setup when NSE changed stock options style from American to European on the recommendation of Securities and Exchange Board of India (SEBI), to attract liquidity to stock options market. We examine both styles of stock options empirically and report interesting observations.

Our results indicate the significant presence of volatility smirk in the individual stock options in the Indian stock options market. Unlike American style stock options, the volatility smirk is a significant predictor of cross sectional future stock returns for the European style stock options which is robust to controls of various firm specific variables such as trading volume and open interest. Further, firms with flatter volatility smirks outperform the firms with

³ For example, see Chan et al. (2002), Mayhew and Stivers (2003), Chakravarty et al. (2004), Pan and Poteshman (2006), Ni et al. (2008), Taylor et al. (2010).

steepest volatility smirks even to control of Carhart four factors. Our portfolio results are robust using non-parametric chi-square test. Results are significant for trading strategies and portfolio management activities as picking stocks with lowest volatility smirks is found optimizing the portfolio returns. A long-short portfolio strategy based on ranking of smirks generates riskless profits for traders and fund managers. Moreover, exploring the origin of anomalies such as volatility smirk might help the researchers and practitioners to develop better options pricing models.

The remainder of this paper is organized as follows. Next section describes the literature surveyed for this study. Section 3 describes the data and methodology and Section 4 presents the empirical results. Section 5 provides the robustness results and finally Section 6 concludes the paper.

2. LITERATURE REVIEW

The Black-Scholes (1973) formula of options pricing states that all options on the same underlying having same time to expiration, but different exercise prices, should exhibit the same implied volatility. However, current empirical evidences show that the implied volatility of the OTM put options is higher than that of the ATM call options (Rubinstein, 1994; and Dumas, Fleming and Whaley, 1998). This phenomenon is known as implied volatility smirk in the option pricing and derivatives literature (Kim and Zhang, 2013). Bates (2000) states that volatility smirks indicate the subjective assessments of investors for future crash probabilities. Bollen and Whaley (2004) and Gârleanu et al. (2009) show that the higher demand for OTM put options contributes to the volatility smirk.

Xing et al. (2010), using individual stock options, document a positive volatility smirk with a median of around 5%. They further examine the predicting ability of volatility smirk for future stock returns and report that stocks with flatter smirks outperform those with steepest smirks, on a risk-adjusted basis. The predictability is robust even after controlling for various cross sectional effects. Yan (2011) examine the predicting ability of implied volatility smile for future stock returns using a sample of four thousand stocks during the period of 1996–2005. He reports a negative relationship between the slope of implied volatility smile and stock returns. The difference between the risk-adjusted average returns of the lowest and highest quintile portfolios is found to be 1.9% per month. Bianconi et al., (2015) use Black-Scholes options pricing model to compute implied volatility and implied risk free rate to re-

price option contracts. They find volatility smile helpful in explaining the differences between market price and model based options price.

However, there are evidences in contrary where volatility smirk is not found to be a robust predictor of future returns (Heston, 1993; Conrad et al., 2007). Heston (1993), for instance, develops an option pricing model with stochastic volatility assuming perfect information flow between the stock market and the options market. The volatility smirks generated using this model fails to predict underlying stock returns, which is attributed to the irrelevancy of expected returns for option pricing.

In related string of literature, Mayhew and Stivers (2003) study the information content of implied volatility about firm level volatility and report that for most actively traded options, the implied volatility subsumes almost all information about firm level volatility. Chakravarty et al. (2004) report that the options market's contribution to price discovery is about 17%, on average. Pan and Poteshman (2006) show that put-call ratios (PCRs) by newly initiated trades have significant predictability for equity returns whereas, Ni et al. (2008) report significant positive relationship between options non market maker demand for volatility and subsequent realized volatility. Taylor et al., (2010) measure the volatility information content of stock options for 149 US firms and the S&P 100 index. They report ATM implied volatilities generally outperforming other measures of volatility forecasts (defined by historical stock returns, ATM implied volatilities and model-free volatility expectations) for every firm. Pathak et al. (2015) examine the information content of options market using S&P CNX Nifty Index options traded on NSE across options moneyness classes and market trends. They report informed traders using ATM options to optimize their portfolio returns.

To the best of our knowledge, the existing literature is silent on whether a particular style of option plays significant role in determining the predicting ability of volatility smirk and its information content about future stock returns. Our study attempts to address this gap.

3. DATA AND METHODOLOGY

3.1 Data

Our sample period of study is from January 01, 2004 to December 31, 2014. We collect end of the day options prices and trading data from NSE daily bhav copy of derivatives market available on NSE website (www.nseindia.com). The spot prices data is collected from prowess database of centre for monitoring Indian Economy (CMIE) and the data for

analysing risk-adjusted returns (i.e. Carhart four factors daily data) is collected from the Indian market data library of Agrawalla, Jacob and Varma (2014)⁴. We classify our sample into two sub-periods, that is, from January 01, 2004 to December 31, 2010 (American period – duration when traded stock options on NSE were of American style); and from January 01, 2011 to December 31, 2014 (European period – during which the stock options traded are of European style). We compute daily volatility smirk following Xing et al. (2010) where the smirk measure ‘ $smirk_{i,t}$ ’ for firm i on day t is the difference between implied volatilities of OTM puts and ATM calls denoted as $Vol_{i,t}^{OTMP}$ and $Vol_{i,t}^{ATMC}$. The implied volatilities are computed by reverting Black and Scholes (1973) options pricing model for volatility.

$$smirk_{i,t} = Vol_{i,t}^{OTMP} - Vol_{i,t}^{ATMC} \quad (1)$$

We classify a put option as OTM if the strike price is between 80 to 95 percent of spot price. An ATM call options has the ratio of strike price to stock price between 0.95 to 1.05. NSE issues stock options contracts for 3 month trading cycle – the near month (one-month), the next month (two-month) and the far month (three-month). We consider options with all maturities in our analysis to avoid missing on any important information, however; we drop the data for options contracts having zero trading volume for a particular day.

Our final sample consists of 29865 firm-day observations during American period and 91644 firm-day observations during European period. The substantially high number of observations in latter period of study, though relatively shorter period; is due to active options trading taking place in many stocks after change in options style from American to European⁵. This signifies the success of NSE in attracting liquidity to stock options market through change in option style and thereby reducing the early exercise risk of an option seller. As we mostly have multiple OTM and ATM options on a stock on any given day, we use a volume weighted approach to arrive at one smirk observations for each firm per day. We use number of contracts traded as weights to compute weighted implied volatility for a firm i on a given day t . Our study period covers 10 years of data during which the number of stocks traded in F&O sections have increased remarkably. Thus, the number of firms has not been the same across the period of study. The minimum and maximum number of firms in our

⁴ They compute factors for Indian market since 1995 using data from CMIE prowest database. More details about the database is available in Agrawalla, Jacob and Varma (2014).

⁵ The share of stock options trading in derivative segment has increased to 6.1% of total turnover during 2013-14 from 2.87% during 2009-10 (share after 9 years of introduction of stock options in Indian market) as reported in Indian Securities market Review (ISMR) published by NSE.

sample has been 17 and 135 respectively. We provide the summary statistics of sample stocks and options over the cross section and then average over the daily time series in Table 1.

Table 1
Summary Statistics for the American and European Options

	IVC	IVP	Smirk	RET	VolC	TOC	OIC	VolP	TOP	OIP
<i>Panel A: American Options Period (January 01, 2004 to December 31, 2010)</i>										
Mean	0.505	0.685	0.180	-0.002	364.2	138.9	12,11,538	47.27	17.87	1,59,402
SD	0.316	0.758	0.720	0.099	753.2	300	22,89,139	128.4	49.79	3,98,830
P50	0.456	0.479	0.027	0.000	115	37.05	4,35,200	9.00	2.94	32,500
Kurt	1306	49.96	79.09	384.9	55.51	5.344	37.76	93.10	9.931	120.3
P25	0.365	0.309	-0.115	-0.016	37	11.05	1,28,000	3.00	.815	7,800
P75	0.579	0.780	0.255	0.018	350	125.9	13,18,000	33.0	11.57	1,30,800
Skew	24.23	5.336	3.692	-16.37	5.795	.5693	4.895	7.513	.7611	7.483
<i>Panel B: European Options Period (January 01, 2011 to December 31, 2014)</i>										
Mean	0.427	0.618	0.191	-0.001	1591	509.9	24,17,337	336.7	103.5	3,93,655
SD	0.161	0.552	0.489	0.060	4758	1542.5	48,54,455	1445	438.1	9,41,695
P50	0.398	0.469	0.068	0	392	123.7	8,00,000	52.0	16.36	1,04,000
Kurt	17.79	39.72	51.34	2808	743	99.11	38.843	6741	862.7	96.63
P25	0.318	0.317	-0.051	-0.014	103	30.87	2,20,000	11.0	3.349	24500
P75	0.499	0.718	0.266	0.013	1378	444.6	23,98,000	223	71.10	3,59,000
Skew	2.208	4.306	4.798	-40.01	19.23	2.246	5.022	54.67	6.226	7.631

Note: Table 1 presents the basic statistics of variables under consideration for the study, which are as follows. Here IVC and IVP are implied volatilities for ATM call and OTM put options respectively. Smirk is the difference between Put and call implied volatilities computed as in equation 1. RET is the daily stock returns. VolC denotes call volume i.e. number of contracts traded of ATM calls and TOC is the turnover of such traded calls in millions Rs. Similarly, VolP and TOP denote the volume (no. of contracts traded) and turnover (in million Rs.) respectively of put options. OIC and OIP are open interests (no. of contracts) of ATM calls and OTM puts respectively. Panel A reports stats on American options traded on NSE during January, 2004-December, 2010 of our study period. Panel B reports the summary stats for the period January, 2011-December 2014, during which the stock options traded on NSE are of European style. The statistics reported are mean, standard deviation (S.D), Median (p50), kurtosis (Kurt), 25th and 75th percentiles (p25 and p75 respectively), and skewness (Skew).

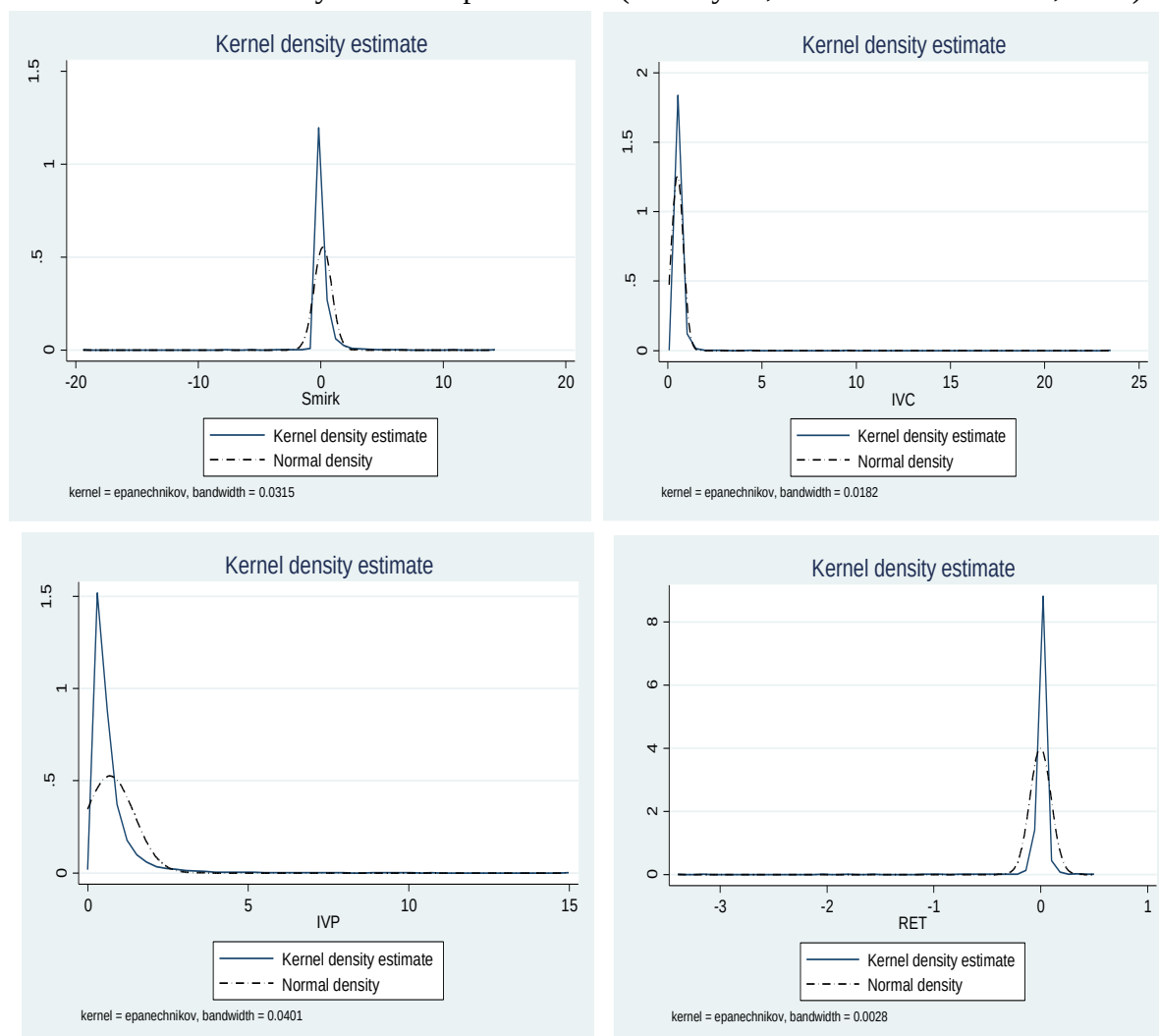
Table 1 suggests that the mean smirk value is positive for both American and European style options indicating that on average, the volatility estimates of put options are higher than that of call options. Similarly, the standard deviations of volatility estimates for put options are higher than call options for both American and European options. However, the volatilities of both call and put options and their variability have reduced post the adoption of European stock options which designate more stability in stock options market. The median volatility smirk of American options is found to be 2.7%, which is approximately 4% lower than the median volatility smirk of European options. The average number of contracts traded for call options are remarkably higher than that of put options during both American and European

options periods and is consistent for both the turnover and the open interests. In line with the expectations of exchanges, the mean volume and turnover of both call and put options have increased many folds after the adoption of European style stock options compared to that of American stock options.

Figure 1 shows the Kernel density plots for key variables, which is a non-parametric way to estimate the probability density function of a random variable. We use the Epanechnikov Kernel function among others, as it is optimal in a mean square error sense. We plot the implied volatilities of call and put options, volatility smirk and daily returns for both American and European style options. The plots confirm the deviation of variables from normality as also indicated by summary statistics.

Figure 1
Kernel Density Plots

Panel A: American Style Stock Options Plots (January 01, 2004 to December 31, 2010)



Panel B: European Style Stock Options Plots (January 01, 2011 to December 31, 2014)

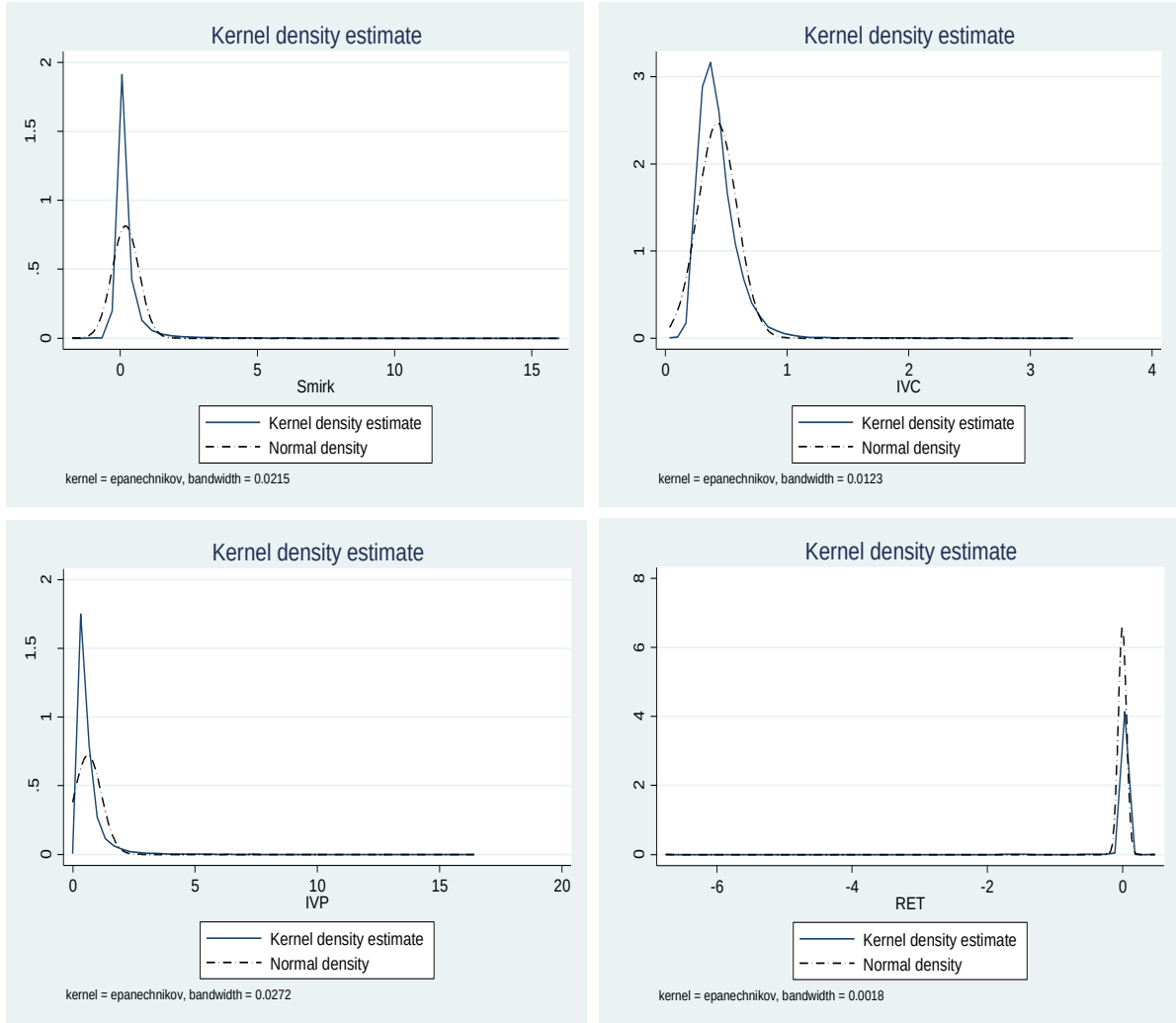


Figure shows the density plots of call and put implied volatilities(IVC & IVP), their Smirks and Returns (Ret) during American and European options periods in Panel A and B respectively. The density plots of the variables are generated using Gaussian function and the standard normal density plots are shown jointly.

3.2 Methodology

First, we examine the predictability of volatility smirk for future stock returns. The volatility smirk reflects the investors' expectation of downward price jump (Xing et al., 2010). If informed traders prefer the options market to place their first trade and the spot market is sluggish to incorporate the implicit information in the options market; we expect the volatility smirk from options market to predict the future stock returns. We run the following Fama–MacBeth (1973), hereafter FM, regression after controlling for potential firm specific trading variables to test whether the volatility smirk predicts the next day returns.

$$Ret_{i,t} = \beta_{0t} + \beta_{1t}smirk_{i,t-1} + \beta_{2t}controls_{i,t-1} + \varepsilon_{it} \quad (2)$$

Where $Ret_{i,t}$ is the returns on stock i on day t , $smirk_{i,t-1}$ is the volatility smirk for firm i on day $(t-1)$ and $controls_{i,t-1}$ is the set of control variables for firm i observed on day $(t-1)$.

With FM regression not only can we examine the significance of the predictability of smirk, but also control for numerous firm characteristics at the same time (Xing et al., 2010). The standard FM regression has two stages. First we estimate the regression in cross section for each day t , and second, we obtain the time series of slope coefficients (β_{1t} and β_{2t}) and investigate their time series assuming they are identically and independently distributed (i.i.d). We report the results of equation (2) in Table 2.

Next, we examine the information content of volatility smirk using portfolio approach. In contrast to the FM approach, the portfolio sorting approach does not impose a condition of linear relation between the interest variable and the return (Xing et al., 2010). Besides, grouping the individual firms into portfolios, the firm-level noise in the data is largely reduced. We apply the portfolio approach in the following manner. For every stock i , $smirk_{i,t}$ is computed on each day (t) and are ranked from lowest to highest. Quintiles are formed each day based on the ranking of $smirk_{i,t}$ where quintile 1 represents portfolio 0 (Low) which consists of 20% of the stocks with the lowest values of $smirk_{i,t}$, whereas quintile 5 or portfolio 4 (High) consists of 20% of sample stocks with the highest $smirk_{i,t}$. The number of stocks in a portfolio is almost equal on each day, however; the number of stocks in a portfolio varies from 3 to 27 as more number of stocks has been included in F&O segment over the sample period. If $smirk_{i,t}$ is informative about future stock returns, the portfolio consisting of stocks with the highest volatility smirk should have minimum returns and vice versa. We follow the trading rule as mentioned below for the portfolios returns analysis:

- i. Compute $smirk_{i,t}$ for each stock using equation (1).
- ii. Buy stocks in the p th quintile ($p=1,...,5$) ranked by $smirk_{i,t}$ at the close price of day t .
- iii. Sell the portfolios at close prices of day $(t+1)$

To compute the average portfolio returns over the period of study, it becomes important to consider that the number of stocks in a particular portfolio vary over time. Manaster and Rendleman (1982) mention that the changing number of stocks over time makes the portfolio returns heteroscedastic because the returns of the portfolio with fewer stocks would be more

volatile than that of portfolio containing large number of stocks. They suggest computing the average portfolio returns by weighting the daily portfolio returns with the number of stocks in the portfolio each day. We use the following expression for the computation of average portfolio returns over the period of study.

$$\overline{R_p} = \frac{\sum_{t=1}^n (W_t)(R_{pt})}{\sum_{t=1}^n W_t} \quad (3)$$

where, $\overline{R_p}$ is the time series average of returns of portfolio p , n is number of days in the study, W_t is the number of stocks included in portfolio p on day t and R_{pt} is the return of portfolio p on day t . The computation of mean portfolio returns using above expression is similar to generalized least square regression estimates where variance of the residual term is supposed to vary with the reciprocal of W_t (Manaster and Rendleman, 1982).

Next, we adjust for risk by applying the following Carhart (1997) four factors model.

$$R_{pt} = \alpha_p + \beta_p MKT_t + \gamma_p SMB_t + \delta_p HML_t + \pi_p WML_t + \varepsilon_{pt} \quad (4)$$

R_{pt} is the excess return of portfolio p on day t , MKT_t , SMB_t , HML_t and WML_t are Carhart(1997) factors⁶. MKT (Market Factor) represents excess return $(r_m - r_f)$ ⁷ on broad market portfolio, SMB (Small minus Big) represents the difference between returns on the portfolio of large stocks and portfolio of small stocks whereas HML (High minus Low) represents the difference between the portfolio returns of high book to market stocks (value stocks) and the low book to market stocks (growth stocks). WML (winners minus losers) factor is computed by subtracting the average returns of the lowest performing firms from the highest performing firms. We expect α_p to be significant for $smirk_{i,t}$ to contain incremental information.

If $smirk_{i,t}$ explains portfolio returns then α_p across portfolios should decrease with portfolio ranks and be significantly different from each other. We test for this phenomenon using long–short portfolio strategy where an excess return on highest smirk portfolio is deducted from

⁶ Fama and French (1992) observed that two classes of stocks have tended to do better than the market as a whole as explained by CAPM (Capital Asset Pricing Model of Sharpe and Linter): They are (i) small caps and (ii) stocks with a high book-to-market ratio (BtM, customarily called value stocks, contrasted with growth stocks. They include two additional factors HML (high minus low) and SMB (small minus big) other than market factor in explaining returns. Carhart (1997) extended three-factor model by adding factor for momentum (WML) computed by subtracting the average returns of highest performing firms from lowest performing firms.

⁷ In the expression $(r_m - r_f)$; r_m represent return on the market portfolio whereas r_f indicates the risk free rate.

the excess return on lowest smirk portfolio. We estimate the following equation for this purpose:

$$R'_{pt} = \alpha'_p + \beta'_p MKT_t + \gamma'_p SMB_t + \delta'_p HML_t + \pi'_p WML_t + \varepsilon'_{pt} \quad (5)$$

where R'_{pt} is the return of long short portfolio investment strategy on day t . We expect α'_p to be positive and significant if smirk is a priced risk factor.

4. EMPIRICAL RESULTS

4.1 Predictability of Volatility Smirk

Table 2 reports the results of FM regression for both American and European options in Panel A and B respectively. The estimates of volatility smirk in predicting next day's stock returns are found to be insignificant when American style stock options traded on NSE. To separate the predicting power of volatility smirk from idiosyncratic characteristics of the firm, we include set of trading variables as controls in the second step. We control for volume of ATM calls and OTM puts due to the linkage between options volume and stock returns established in literature (Schlag and Stoll, 2005; Pan and Poteshman, 2006). We also control for open interest, as it is an important measure of options trading activity and represents a useful proxy of divergence in trader's belief (Bessembinder et al., 1996). Next, the total turnover is used as proxy to control for firm size in the cross section. Numerous studies show that size is an important factor in determining stock returns (Xing et al. 2010). We run the second step regression including all the controls and observe similar results except turnover affecting the returns, which substantiates the size effect.

However, we observe contrasting results after the adoption of European style stock options by NSE. The first step regression results indicate that the volatility smirk becomes a very significant predictor of stock returns when European style stock options began trading. The regression coefficient is found to be -0.012 which is statistically significant at 1% level of significance. Moreover, the negative sign of the co-efficient is consistent with the literature that volatility smirk is indicative of downside price movement. In the second step regression, we find that the inclusion of control variables have not reduced the predictive power of volatility smirk, rather it becomes even more prominent. The smirk co-efficient changes to -0.019 with a t -statistic of -11.56 ; which is an improvement in both magnitude and statistical significance of the predictability.

Table 2

Estimates of Fama–MacBeth Two Step Regression

		Smirk _{i,t-1}	VolC _{i,t-1}	OIC _{i,t-1}	VolP _{i,t-1}	OIP _{i,t-1}	TO _{i,t-1}	R ²
<i>Panel A: American Options Period (January01, 2004 to December 31, 2010)</i>								
I	Coeff	−0.0049						0.0892
	t-stat	(−1.03)						
II	Coeff	−0.0136	0.0024	0.0017	−0.0065	−0.0036	0.039***	0.3869
	t-stat	(−1.21)	(0.44)	(0.31)	(−1.43)	(−0.63)	(3.44)	
<i>Panel B: European Options Period (January01, 2011 to December 31, 2014)</i>								
I	Coeff	−0.012***						0.0452
	t-stat	(−7.76)						
II	Coeff	−0.019***	0.009***	−0.002***	−0.008***	−0.0077	−0.0016	0.2727
	t-stat	(−11.56)	(8.38)	(−2.60)	(−11.99)	(−1.20)	(−1.30)	

Note: Table 2 presents Fama-MacBeth two-step regression estimates. The first step has lag smirk as the only regressor whereas step two has set of control variables along with Smirk. VolC_{t-1} and VolP_{t-1} are the volume of ATM calls and OTM puts measured by log of total number of such contracts traded on previous day. OIC_{t-1} and OIP_{t-1} are the log of open interests on ATM calls and OTM puts respectively as on day t-1. TO_{t-1} is the log of total turnover of ATM calls and OTM puts on the previous day. The t-statistics of the co-efficients are reported in parenthesis. The R-squared of the regressions are reported in the last column.

** and *** indicates significance at 5% and 1% levels.

Besides, volume for the call options are found to be affecting the stock returns positively as opposed to negative effect of put options volume, which is consistent with the argument that increase in volume of call and put options is indicative of good and bad news respectively. Interestingly, we do not find a significant co-efficient for turnover. We believe that the surge in trading volume of stock options after the adoption of European style has reduced the cross sectional size differences among stocks in F&O segment and therefore, turnover no more remains an important determinant of future returns. The call open interests affect the returns negatively. This is possible when traders with negative news open one-sided position through selling a call option, which suppresses the stock prices. Unlike American options, such trading activities become less risky for sellers of European options pertaining to risk of premature exercise.

Our FM regression results signify the profit making opportunities arising for traders through short selling of stocks post the adoption of European style stock options. The results justify the recommendation of SEBI to change the style of stock options offered by the exchanges in India, and the adoption of the same by NSE, to attract liquidity in Indian stock options market.

4.2 Testing Information Content of Volatility Smirk using Portfolio Approach

In this section, we examine the information content of volatility smirk using portfolio approach. First, we rank the sample firms based on volatility smirk on daily basis and form quintiles. Here, the first quintile (portfolio 0) contains the lowest smirk stocks and the last quintile (portfolio 4) contains firms with the highest smirk values. Then we compute the portfolio returns following the trading rule. Table 3 presents the average portfolio returns computed using equation (3) over the period of study. Our results indicate that the portfolio returns decreases going from lower to higher ranked portfolios for both American and European options, which is consistent with the expectation.

Table 3
Portfolio Returns Based on Ranking of Volatility Smirk

Ranks for Smirk		Low (0)	1	2	3	High (4)
American	Returns	0.00241	0.00075	−0.00244*	−0.00462***	−0.00797***
	(t-stat)	(1.096)	(0.533)	(−1.901)	(−3.981)	(−4.891)
European	Return	0.00520***	0.00144**	−0.00188**	−0.00420***	−0.00585***
	(t-stat)	(7.433)	(2.346)	(−2.535)	(−6.917)	(−6.595)

Note: Table 3 shows the returns (%) of portfolios formed based on the ranking of volatility smirk. Portfolio 4 contains firms with most pronounced smirk values and vice-versa for portfolio 0. We report results separately for periods when American and European styled stock options traded on NSE. Results include the average portfolio returns over the period of study along with the corresponding t-statistics reported in the parenthesis. *, ** and *** indicates significance at 10%, 5% and 1% levels respectively.

However, the returns of portfolios with flatter volatility smirk firms are not found significantly different from zero in the case of American options. This can be attributed to the dry liquidity in stock options market before 2011. The results are more pronounced in case of European options which again rationalize the change of options style in an attempt to attract traders. It provides a trader average daily returns of 0.11% if long in portfolio 0 and short in portfolio 4.

Next, we adjust for risk by applying the Carhart four factors model to examine if volatility smirk provide returns over and above the popular Carhart four factors that affect returns. We present the regression results in Table 4. For volatility smirk to be an additional factor determining stock returns, we expect the intercept (α_p) to be significant in the regressions across portfolios. We observe the expected results consistent for both styles of stock options. Moreover, we also expect α_p to be decreasing with increase in portfolio ranks for theoretical justification of volatility smirk being indicative of downward movement in prices. We find

α_p to be decreasing from -0.024% to -0.032% for American options and -0.028% to -0.036% for European options as we go from low smirk to high smirk portfolio however, for American options the trend is not clear.

Table 4
Carhart Four Factors Regression Results

	α_p	MKT	SMB	HML	WML	R ²
<i>Panel A: American Options Period (January 01, 2004 to December 31, 2010)</i>						
Low (0)	−0.024*** (−6.84)	0.007*** (2.97)	−0.003 (−0.43)	−0.000 (−0.02)	0.007 (1.36)	0.022
1	−0.030*** (−8.94)	0.010*** (4.12)	0.005 (0.86)	−0.001 (−0.23)	0.008* (1.86)	0.030
2	−0.038*** (−6.98)	0.012*** (3.20)	0.013 (1.37)	−0.006 (−0.68)	−0.005 (−0.64)	0.012
3	−0.032*** (−10.83)	0.009*** (4.59)	0.001 (0.02)	0.006 (1.25)	0.002 (0.53)	0.035
High (4)	−0.032*** (−119.3)	0.014*** (69.20)	0.003*** (5.62)	0.001*** (2.60)	−0.002*** (−4.12)	0.187
Low–High	0.009*** (2.84)	−0.008*** (−3.38)	−0.006 (−1.10)	0.000 (0.08)	0.010** (2.19)	0.013
<i>Panel B: European Options Period (January 01, 2011 to December 31, 2014)</i>						
Low (0)	−0.028*** (−29.95)	0.009*** (7.92)	−0.000 (−0.65)	0.000 (0.36)	−0.002* (−1.66)	0.121
1	−0.031*** (−32.43)	0.011*** (8.44)	−0.003* (−1.80)	0.001 (0.14)	−0.001 (−0.39)	0.128
2	−0.035*** (−28.55)	0.012*** (7.68)	−0.001 (−0.54)	−0.001 (−0.43)	−0.001 (−0.85)	0.098
3	−0.036*** (−89.52)	0.014*** (26.28)	−0.0001 (−0.21)	0.002*** (4.25)	−0.001 (−1.06)	0.575
High (4)	−0.038*** (−386.0)	0.017*** (126.3)	0.000 (5.14)	0.003*** (26.70)	−0.000** (−1.97)	0.257
Low–High	0.011*** (10.11)	−0.006*** (−4.49)	−0.001 (−0.35)	−0.002 (−1.22)	−0.002 (−1.14)	0.028

Note: Table 4 presents the risk-adjusted returns of portfolios for both American and European stock options in panel A and B respectively. Here portfolio 0 (Low) contains the stocks with lowest volatility smirks. Portfolio 4 (High) has stocks with steepest smirk. We report the factor coefficients of equation 4 and their respective t-statistic in parenthesis, for each portfolio. The (Low-High) shows the factor coefficients of long-short portfolio strategy as in equation 5. The R² of the regressions are shown in the last column. EMR, SMB, HML and WML are Carhart four factors as explained in the text. *, ** and *** indicates significance at 10%, 5% and 1% levels.

We find portfolio 3 earns lowest returns (-0.038%) instead of portfolio 4. Nonetheless, we find a very consistent pattern of α_p for European options. Among other factors, excess market returns (MKT) is found to be consistently significant in both panel A and B results.

This is consistent with the celebrated CAPM model of asset pricing and signifies that market factor is an important determinant for future returns. We observe the other factors to be significant predictor only for high rank portfolio implying that size, value and momentum premiums are not available for stocks in lower quintiles. The R^2 of the regressions have improved substantially during the period when European style stock options traded on NSE.

The long short portfolio strategy results using equation (5) are shown in the row Low–High of Table 4. We expect the intercept α'_p to be positive in this regression for a riskless profit to traders. Our results indicate a daily riskless profit of 0.009% (or 3.28% annualized) with a t–statistic of 2.84 and 0.011% (or 4.01% annualized) with a t–statistic of 10.11, for American and European options respectively. We do not find factors other than excess market returns as a significant predictor of portfolio returns in long–short portfolio strategy. However, all the risk factors i.e. size (SMB), value (HML) and momentum (WML) are observed to be significant predictor of future stock returns for steepest smirk portfolio for both American and European options.

Overall, we find the firms with flatter volatility smirk over–perform firms with steepest volatility smirk on the next day and the returns differences are economically high and statistically significant. We observe the results to be more pronounced when European style stock options started trading on NSE. We attribute the results to the reduced prematurity exercise risk of options seller in European options. This characteristic possibly has attracted informed traders with negative information to the stock options market resulting in profitable opportunities due to delay of spot market in incorporating this information.

5. ROBUSTNESS TEST

To test for robustness of our results, we use non–parametric chi–square (χ^2) test similar to Manaster and Rendleman (1982) on returns to examine if the portfolio formation based on ranking of volatility smirk is different from random portfolio selection. To perform this test, we rank the returns of the five portfolios on each day. We argue that if portfolio formations are completely random, the returns rankings on the five portfolios should not be related to the volatility smirk rankings. On the other hand, if volatility smirk provides information, the higher ranked portfolios should earn lesser returns and vice–versa. In Table 5, each element of the 5×5 matrix represents the number of days out of total days of the study (884 days and 978 days for American and European options respectively) that a particular portfolio had the

indicated return rank. That the number of days is lesser during the American options trading period on NSE is due to the sparse trading in stock options during the initial period. The χ^2 statistic is used to determine if the pattern of numbers in the matrix is different from random pattern.

Table 5
Frequency Distribution of Return Ranks of Volatility Smirk Based Portfolios

<i>Panel A: American Options Period (January 01, 2004 to December 31, 2010)</i>						
	Low	1	2	3	High	Total
Low	99	99	171	200	315	884
1	99	155	193	265	172	884
2	148	201	208	179	148	884
3	222	232	188	148	94	884
High	316	197	124	92	155	884
Total	884	884	884	884	884	884
$\chi^2 (16) = 526.4539^{***}$						

<i>Panel B: European Options Period (January 01, 2011 to December 31, 2014)</i>						
	Low	1	2	3	High	Total
Low	58	98	133	309	380	978
1	48	116	261	332	221	978
2	90	232	317	175	164	978
3	215	334	192	106	131	978
High	567	198	75	56	82	978
Total	978	978	978	978	978	978
$\chi^2 (16) = 1923.0777^{***}$						

Note: Table 5 presents the frequency distribution of portfolios based on ranking of returns for both American and European options in Panel A and B respectively. The no-information hypothesis upholds that the frequency in each of the cells above should be equal to 884/5 and 978/5 for American and European periods respectively. The chi-square test statistics are shown with degree of freedom in parenthesis. *** indicates significance of statistics at 1% level.

The results of the χ^2 test are encouraging for both American and European options. The value of $\chi^2(16)$ is found to be high and statistically significant at 1% level for both styles of options validating the probability that the observed value in the matrix could be realized purely on chance is nearly zero. The observed distribution suggests that returns on the portfolio are largely as expected. The observations are found to be clustering on the minor diagonal of the matrix, which suggest that portfolio containing stocks with smallest smirk values earns highest returns for maximum number of days and vice-versa.

Interestingly, we notice that the χ^2 statistic remains significant for American options though the returns are significant only for higher ranked portfolios (see Table 3). One possible

reason for χ^2 statistic remaining significant despite t -statistic indicating insignificant portfolio returns is the insensitivity of χ^2 statistic towards dispersion in portfolio returns. For example, a very high return earned unexpectedly by a higher ranked portfolio on few days would increase the variance of that portfolio returns. This increases the standard error which renders the t -statistic to be very low and insignificant. The χ^2 test results signify the possible informational role of volatility smirk in determining future stock returns.

6. CONCLUSIONS

Informed traders can decide on trading venue based on enhancement of trading incentives on their informational advantages. As a result, lead-lag relationship in the price discovery process exists in markets where homogeneous or closely linked securities trade. In this paper, we investigate the pattern of volatility smirk and its predictability and information content for the underlying stock's future returns. The study becomes distinctive as we examine a unique market set up where the styles of options have been changed by the exchange from American to European to attract liquidity to the stock options market. We define the volatility smirk variable as the difference between the implied volatilities of OTM puts and ATM calls.

We observe the mean smirk values to be positive for both American and European options indicating that on average, put options, which are vehicle to trade on negative news, estimates volatility on higher side. We document that the volatility smirk computed on European options (unlike American options) can predict the underlying cross-sectional equity returns even after adjusting for controls of various firm specific variables such as trading volume and open interest. Moreover, firms with steepest volatility smirk underperform firms having a flatter volatility smirk which is robust to the control of the Carhart four factors. This indicates that volatility smirk is a priced risk factor determining future stock returns. The non-parametric chi-square test indicates that portfolio returns results are robust. We argue that the informational advantage of option traders might be the reason for the observed predictability.

These results indicate the potential of profit making for traders and fund managers based on implied volatility estimates provided by series of options trading on a stock. Further, we observe that the informed traders with negative news have been attracted even more towards stock options (possibly by selling calls or buying puts) market post the change in stock option style from American to European. This is evident from an increase in share of stock options in total turnover of derivatives market on NSE. The reduced pre-maturity exercise risk in selling of European options might have resulted in such a preference of stock options for

traders in Indian market. More pronounced empirical results for predictability and information content of volatility smirk for European options validate our argument of trader's preference in these options. Nonetheless, going European seems to be a successful attempt of NSE to enhance liquidity of stock options market.

REFERENCES:

- Agarwalla, S. K., Jacob, J., & Varma, J. R. (2014). Four factor model in Indian equities market. *Indian Institute of Management, Ahmedabad Working Paper*, (2013–09), 05.
- Bates, D. S. (1991). The crash of '87: Was it expected? The evidence from options markets. *The journal of finance*, 46(3), 1009–1044.
- Bates, D. S. (2003). Empirical option pricing: A retrospection. *Journal of Econometrics*, 116(1), 387–404.
- Bates, S.D. (2000). Post-'87 crash fears in the S&P 500 futures option market. *Journal of Econometrics*, 94, 181–238
- Bessembinder, H., Chan, K., & Seguin, P. J. (1996). An empirical examination of information, differences of opinion, and trading activity. *Journal of Financial Economics*, 40(1), 105–134.
- Bianconi, M., MacLachlan, S., & Sammon, M. (2015). Implied volatility and the risk-free rate of return in options markets. *The North American Journal of Economics and Finance*, 31, 1–2
- Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *The journal of political economy*, 637–654.
- Bollen, N. P., & Whaley, R. E. (2004). Does net buying pressure affect the shape of implied volatility functions?. *The Journal of Finance*, 59(2), 711–753.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1), 57–82
- Chakravarty, S., Gulen, H., & Mayhew, S. (2004). Informed trading in stock and option markets. *The Journal of Finance*, 59(3), 1235–1258.
- Chan, K., Chung, Y. P., & Fong, W. M. (2002). The informational role of stock and option volume. *Review of Financial Studies*, 15(4), 1049–1075.
- Chen, C. R., Lung, P. P., & Tay, N. S. (2005). Information flow between the stock and option markets: Where do informed traders trade?. *Review of Financial Economics*, 14(1), 1–23.

- Chen, S. X., & Xu, Z. (2014). On implied volatility for options—Some reasons to smile and more to correct. *Journal of Econometrics*, 179(1), 1–15.
- Conrad, J., Dittmar, R. F., & Ghysels, E. (2008). Skewness and the Bubble. *Unpublished working paper, University of North Carolina at Chapel Hill*.
- Cummings, J. R., & Frino, A. (2011). Index arbitrage and the pricing relationship between Australian stock index futures and their underlying shares. *Accounting & Finance*, 51(3), 661–683.
- De Jong, F., & Donders, M. W. (1998). Intraday lead–lag relationships between the futures–, options and stock market. *European Finance Review*, 1(3), 337–359.
- Dumas, B., Fleming, J., and Whaley, R.E. (1998). Implied volatility functions: Empirical tests. *The Journal of Finance*, 53, 2059
- Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *the Journal of Finance*, 47(2), 427–465.
- Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *The Journal of Political Economy*, 607–636.
- Foresi, S., & Wu, L. (2005). Crash–o–phobia: A domestic fear or a worldwide concern?. *The Journal of Derivatives*, 13(2), 8–21.
- Gârleanu, N.; L. H. Pedersen; and A. Potesman. “Demand–Based Option Pricing.” Working Paper, University of Pennsylvania (2007).
- Heston, S. L. (1993). A closed–form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.
- Kim, J. B., & Zhang, L. (2014). Financial reporting opacity and expected crash risk: Evidence from implied volatility smirks. *Contemporary Accounting Research*, 31(3), 851–875.
- Kumar, U., & Tse, Y. (2009). Single–stock futures: Evidence from the Indian securities market. *Global Finance Journal*, 20(3), 220–234.
- Manaster, S., & Rendleman, R. J. (1982). Option prices as predictors of equilibrium stock prices. *The Journal of Finance*, 37(4), 1043–1057.
- Mayhew, S., & Stivers, C. (2003). Stock return dynamics, option volume, and the information content of implied volatility. *Journal of Futures Markets*, 23(7), 615–646.
- Ni, S. X., Pan, J., & Potesman, A. M. (2008). Volatility information trading in the option market. *The Journal of Finance*, 63(3), 1059–1091.
- Pan, J. (2002). The jump–risk premia implicit in options: Evidence from an integrated time–series study. *Journal of financial economics*, 63(1), 3–50.

- Pan, J., & Poteshman, A. M. (2006). The information in option volume for future stock prices. *Review of Financial Studies*, 19(3), 871–908.
- Pathak, R., Bhattacharjee, K. & Reddy V. N. (2015). Information Content of Derivatives under Varying Market Conditions and Moneyness: The Case of S&P CNX Nifty Index Options. *Global Business Review*, 16(2), 281–302.
- Rubinstein, M. (1994). Implied Binomial Trees. *The Journal of Finance*, 49, 771–818.
- Schlag, C., & Stoll, H. (2005). Price impacts of options volume. *Journal of Financial Markets*, 8(1), 69–87.
- Taylor, S. J., Yadav, P. K., & Zhang, Y. (2010). The information content of implied volatilities and model-free volatility expectations: Evidence from options written on individual stocks. *Journal of Banking & Finance*, 34(4), 871–881.
- Theissen, E. (2012). Price discovery in spot and futures markets: a reconsideration. *The European Journal of Finance*, 18(10), 969–987.
- Wang, J. (2011). Price behavior of stock index futures: Evidence from the FTSE Xinhua China A50 and h-share index futures markets. *Emerging Markets Finance and Trade*, 47, 61–77.
- Xing, Y., Zhang, X., & Zhao, R. (2010). What does the individual option volatility smirk tell us about future equity returns? *Journal of Financial and Quantitative Analysis* 45(3), 641–662.
- Yan, S. (2011). Jump risk, stock returns, and slope of implied volatility smile. *Journal of Financial Economics*, 99(1), 216–233.